

Meeting 1 – 16/02/2026

Attendees: Swapnil sir, Chilaka, Devadatta, Govardhan

Discussion Points:

- Discussed the overall **project proposal**.
- Discussed the work done on using Regex for identifying people's names as they are in standard format in all docs.
- Sir suggested focusing on **one specific task** due to limited time.
- Suggested task: **Identify ministries involved in a given text**, even when the ministry name is **not explicitly mentioned**.
- Avoid large tasks like:
 - Summarization
 - Sentiment analysis
 - Other complex NLP tasksbecause they require **annotated data and heavy evaluation**.
- Recommendation: **Choose a small task and study it deeply**.

Suggestions from Sir:

We should be able to explain:

- Why did we select this task?
- Why are we using a particular method?
- Why should the method work?
- **When and why the method might fail?**

TODO:

- Narrow the project scope to **ministry identification task**.
- Prepare a **revised proposal focusing on this task**.
- Think about **possible approaches and justification**.

Meeting 2 – 02/03/2026

Attendees: Swapnil sir, Chilaka, Devadatta

Discussion Points:

- Discussed the **updated proposal** focusing on **identifying ministries from text**.
- Explained the planned **methodology**:
 - Use **TF-IDF to identify ministry keywords**.
 - Map text to ministries using:
 - **Keyword matching**
 - **Vector embedding comparison model**.

Suggestions from Sir:

- Experiment with **different numbers of TF-IDF keywords**.
- Analyze:
 - How changing the number of keywords affects **probability distribution across ministries**.
 - Situations where the model **fails to identify the correct ministry**.
- Think about how to **use the prior knowledge** that only a few ministries (out of all 53) will be discussed through a session, **to modify/bias the probabilities?**

Evaluation Discussion:

- Use an **LLM (e.g., ChatGPT)** to label sample documents.
- Label **each paragraph with the ministry involved**.
- Compare **Model predictions vs LLM labels**
- Analyze differences:
 - Why they occur
 - Where the model performs well or poorly.

Documentation Requirement:

Clearly explain in the report, How the **reference data was collected**.

Sources used:

- **Parliament website PDFs** for parliament vocabulary.
- Information about **53 ministries** collected from:
 - Ministry websites
 - Wikipedia
 - LLMs.

TODO:

- Implement **TF-IDF keyword extraction for ministries**.
- Build **ministry knowledge base / vocabulary**.
- Test **different keyword counts**.
- Create a sample **dataset for evaluation**.
- Generate **LLM labels for comparison**.
- Document **data sources and methodology clearly in the report**.

Meeting 3 – 09/03/2026

Attendees: Swapnil sir, Chilaka, Devadatta

Discussion Points:

- Converted **53 ministries into vector embeddings** using **Sentence Transformer**.
- Each ministry is represented as a **384-dimension vector**.
- Visualized ministry vectors using 2D and 3D plots from **t-SNE** and **PCA**.
- The plots show **clear separation between many ministries**, indicating embeddings capture semantic differences.

TF-IDF Vocabulary Analysis:

- TF-IDF generated ministry vocabularies were analyzed using **frequency charts**.
- Observation:
 - Many words are **common across multiple ministries**.
 - These words belong to the **general parliament domain** and are not useful for distinguishing ministries.

Decision:

- Create a **domain-specific stopwords list** to remove common parliament words from TF-IDF vocabularies.

Suggestion from Sir:

- Use **Topic Modelling** to identify:
 - Topics in the documents
 - Important keywords within each topic.
- Concept discussed:
 - **Documents = distribution over topics**
 - **Topics = distribution over words**
- Suggested tool:
Online Topic Modeling Tool by David Mimno

Future Work:

- Later compare **classification performance using different approaches**, such as:
 - TF-IDF keyword based method
 - Embedding similarity method
 - Topic-based approach

TODO:

- Create **domain-specific stopwords list for parliament terms**.
- Clean TF-IDF vocabularies using this stopwords list.
- Experiment with **topic modelling** using David Mimno's tool.
- Extract **topics and keywords** from sample documents.
- Compare **results from different ministry classification approaches**.

Meeting 4 – 16/03/2026

Attendees: Swapnil sir, Chilaka, Devadatta, Govardhan

Discussion Points:

- Compared **classification results** from:
 - Vocabulary (TF-IDF) based method
 - Embedding similarity based method
- Observed that **results from both methods are very different**.
- Also observed that **cosine similarity values for embeddings are almost the same across ministries**, which is not ideal.
- In contrast, **vocabulary-based probabilities are unevenly distributed**, which looks more reasonable.

Suggestion from Sir:

- Verify the assumptions and code.
- Are 30 vocabulary keywords per ministry sufficient?
- Do the embeddings correctly represent each ministry?
- Currently, full ministry text is converted into a single embedding:
 - Is this a valid approach?
 - Will information get diluted or degraded when converting large text into one vector?

Additional Methods to Explore:

- Information Gain for feature selection
- Naive Bayes for classification

Also discussed Topic Modelling:

- Since topic modelling is unsupervised, need to think about:
 - How to **relate topics to ministries**
 - How to **handle independence between topics and ministries**

Future Work:

- Later compare **classification performance using different approaches**, such as:
 - TF-IDF keyword based method
 - Embedding similarity method
 - Topic-based approach

TODO:

- Verify assumptions in current workflow.
- Test **different number of vocabulary keywords** per ministry.
- Investigate whether **full-text embedding is appropriate** or try alternative embedding strategies.
- Study **Information Gain** for feature selection.
- Study **Naive Bayes classifier** for ministry classification.
- Think about **mapping topics to ministries** in the topic modelling approach.
- Compare results from:
 - TF-IDF vocabulary method
 - Embedding similarity method
 - Topic modelling method
 - Naive Bayes / Information Gain based method

Meeting 5 – 30/03/2026

Attendees: Swapnil sir, Chilaka, Devadatta

Discussion Points:

- Refined **paragraph extraction process** after identifying flaws.
- Improved ministry embedding approach:
 - Instead of one vector per ministry,
 - Created **one vector per paragraph of ministry text**,
 - Then computed cosine similarity between paragraph vectors,
 - Used **maximum similarity** for ministry scoring.
- Compared vocab-based and embedding-based classification again:
 - Some ministries matched in both
 - Some mismatched
 - Computed **Top-5 error**
 - Found ministries with **similar domains are clashing**:
 - e.g. Agriculture & Food, AYUSH & Health, Mines & Coal
- Decided to **combine similar ministries** and reduce the number of classes.
- To justify grouping statistically:
 - Performed clustering using:
 - KMeans, GMM, Agglomerative, Spectral Clustering, KNeighbors Graph
 - Due to high semantic overlap, clustering separation was weak (low silhouette score).
 - Adopted **consensus-based (co-occurrence) clustering** to find stable ministry groups.
- Topic modelling using **David Mimno tool**:
 - Generated 25 topics.
 - Checked topic distribution over ministries.
 - Generated embeddings for topics.
 - Computed cosine similarity between topics and ministries.
 - Assigned topics to ministries.
- **Next plan**:
 - Perform classification using **topics + grouped ministries**
 - Compare with original methods.

Sir suggested:

- Instead of mapping **one topic -> one ministry**, allow **one topic -> multiple ministries**.
- This will cover all the 53 ministries instead of only 25.

TODO:

- Finalize ministry grouping.
- Perform topic-based classification.
- Compare:
 - Vocab-based
 - Embedding-based
 - Topic-based
 - Grouped ministry classification
- Allow multiple ministries per topic instead of single mapping.